

Advancing Home Performance House By House

A small home performance company is growing by leaps and bounds by providing something of value to all its customers—including an education in home performance.

by **Leslie Jackson**

As the new associate editor for *Home Energy* magazine, I was assigned to cover a blower door test at the home of the magazine's managing editor, Jim Gunshinan, in Walnut Creek, California. Was this to be another case of the cobbler's shoes, or would the results be unexciting? The house looked pretty sound from the outside, with its moderate size and modern doors and windows. I arrived by bicycle at the house, on a peaceful, tree-lined street, to find Advanced Home Energy's well-stocked biodiesel truck in the driveway. Company President Ori Skloot was directing his two coworkers to prepare the house for a blower door test by using blue painting tape and cardboard to seal the registers and the attic door. All of them had removed their shoes out of respect for the home and this added a refreshing element of polite comfort. Meanwhile, Skloot explained all the steps of a home performance test process to me and answered my questions in lay language. I found the claim that's made on the Advanced Home Energy Web site to be true: "We keep you thoroughly educated and informed every step of the way."

Advanced Home Energy (AHE) seems well on its way to success, with plans to expand its business in the future. AHE is not just a full-service home performance contractor; the

company carries out blower door tests, troubleshoots leaks, and advises on energy efficiency, but it goes further to embody a whole-systems approach. This year to date AHE has done tests and home performance retrofits in over 300 homes.

Success with the Holistic Approach

AHE discovered early on that clients view an average insulation job as a commodity, and will often look for the low-cost bidder. AHE differentiated itself on quality, but nonetheless found itself competing against contractors who would bid for a job and claim that they could do it in a third of the time that it would take AHE to do the same job. Clients who didn't know the difference assumed that all insulation work was the same and would hire the other contractor. Skloot says, "Up until four months ago our average insulation job was \$1,800. When we made the switch to home performance work we found that our average job for that work is \$12,000." Next year AHE expects to decrease its number of clients as it becomes more selective in the homes it serves. "We have become more selective in the jobs we are willing to do—we are now walking away from potential work when we feel that the work the client is



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requesting will not achieve the savings and comfort they are expecting or needing. For example, some people call us wanting attic insulation and when we see their attic we see that there is a lot of air sealing that needs to happen first. We then propose that and explain to them the importance. If they are not willing to seal first



JESSIE JACKSON

Advanced Home Energy's Manuel Matamoros uses a smoke pencil to test for air leaks around a window.

The more homeowners understand the effect that their own house has on their health, on the environment, and on their wallet, the more likely they are to undergo a retrofit.

If AHE is called to consult on a new construction job, it includes all parties involved in a plan for energy-efficient design, in order to maximize these values from the beginning. In this way, AHE helps to design out potential problems in a house's energy consumption before the house is built, which can prevent the kinds of problems that would require retrofitting later on. This first step of the holistic process can be a point of education for architects, homeowners, and building contractors. During the later stages of construction, or during

an energy analysis and retrofit, AHE may design and implement a residential PV system that is properly sized to the house. AHE will always perform testing and remediation work before installing a PV system, so as not to over-design it. (The state of California now requires PV contractors to analyze a home's energy efficiency before installing a PV system; however, implementing the efficiency measures is still optional.) Continuing further along the holistic spectrum, AHE may examine the role of water usage in the house's energy cycle. The company can trace where water is being wasted and recommend simple ways to conserve water, divert storm drain flow, and create landscaping oases. AHE even has dreams of incorporating the waste stream into the whole-house picture, looking at what comes in and what goes out. Reducing in both directions conserves energy, so why not consider

the waste stream part of the house's energy cycle?

Skloot says most homeowners don't need much convincing once they understand all the benefits of following the suggestions of a home performance contractor, whether they do all of the work right away, or invest in steps along the path to comfort, health, and energy efficiency. He says his clients' first priority is comfort. Most contractors say that comfort, health, and energy efficiency are their clients' primary motivators for these upgrades (see "The Smart Sell," p. 30). Appealing to their hearts and common sense is the quickest way to win clients over to their services. Jim Gunshinan, who knows quite well the benefits of a retrofit, was surprised to find out where his investment was going to pay off (see "Leaky Downlights Waste Home Energy").

A Business Model Rooted in Service

Triple bottom line is an accounting principle that views social, ecological, and economic benefits as equally important. People, planet, and profit are integral to the bottom line, not just profit. While his coworkers adjusted the blower door and continued to air seal the interior of Jim's house, I asked Skloot how he got his start, where he thinks the business of home performance is going, and whether he is confident in the triple bottom line as a business strategy. Not surprisingly, the triple bottom line has been on his mind since the early days of his sustainability studies, has been a priority in his ambitions, and is a reality in his business. Always a fan of Mother Nature, he was inspired to learn more about renewable energy while he was in Israel on a research grant from the University of California at Berkeley's Haas School, where he was studying business with an emphasis on sustainability. At the time, environmentalism took a back seat to human survival in Israel, and

we may opt to not do the work at all because we aren't confident that we will make a significant enough impact. We don't want to do work and then have a disappointed client that doesn't notice any change after the work is complete." Nonetheless, the company expects to double its gross and net revenue. It currently employs eight full-time staff and foresees hiring another four people in the next year to help with sales, performance testing, and project management.

We joked together that homes don't come with an owner's manual, or repair records on what lies "under the hood," from electrical wiring to ductwork. For AHE, educating homeowners, who are crucial players in energy conservation, is an integral part of the holistic approach. Residential heating and cooling equipment and water heaters consume the lion's share of all energy used in the U.S. residential sector.

resources were scarce. He saw large black containers full of water on the roofs of all of the houses, holding the sun's energy to deliver hot water to the residents. This had a big impact on him. Seeing how a simple innovation could solve an everyday need got him thinking about renewable energy, available resources, and sustainable business possibilities back at home.

After graduating from Haas, Skloot wasted little time. In the first year, he cofounded Rising Sun Energy Center (RSEC), a Berkeley, California-based nonprofit organization focused on promoting energy awareness, renewable energy, and resource conservation through educational and community service programs. What started as a \$40,000 summer program administered from Skloot's apartment, continues today with a summertime staff of 100 people, a budget of \$2 million, and programs all over the San Francisco Bay Area (see "Training for Service," *HE* Sept/Oct '07, p. 16). While he was executive director at RSEC, Skloot learned about building home performance, and he became a strong advocate after attending a six-day training course run by the California Building Performance Contractors Association (CBPCA). More and more he found himself advising RSEC clients about the benefits of conducting home performance work prior to installing solar PV. People would walk into his office interested in installing a \$30,000 PV system and leave thinking about home comfort and efficiency. It was only a matter of time before Skloot decided that he wanted to educate clients and enjoy the benefits of actually doing the work.

What inspires a contractor or other business professional to invest in the training and tools to pursue home performance contracting? The business of home energy performance contracting is not new. Many existing HVAC and other home heating and cooling professionals add this service to their shingle, and many contractors

Leaky Downlights Waste Home Energy

Ori Skloot of Advanced Home Energy in Berkeley, California, came to my house and took care of my recessed-can problem. American houses, especially the new ones, have a lot of recessed-can lights, also known as downlights, (see "Recessed Lighting in the Limelight," *HE* Jan/Feb '04, p. 12). A study of northeastern homes in the United States found an average of 23 recessed-can lights per home. New California houses have an average of six downlights in their kitchens alone! My house was built without them in 1951, but the previous owners had seven of them installed.

Homeowners pay a heavy price for well-lit kitchens unless those downlights are air sealed and insulated. When Ori and his crew depressurized my house to 50 Pa by closing all the windows, covering all the registers, and installing a blower door in the front door, we found that my house was leaky. Before Ori took care of my leaky can lights, my one-story, 1,170-ft² house measured 11.5 air changes per hour (ACH) at 50 Pa, which translates to approximately 0.72 ACH by natural ventilation. The American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE) recommends a total of 0.35 ACH, provided by natural ventilation or mechanical ventilation, as a good measure for healthy indoor air quality. They also recommend that fresh air come from outside in a controlled fashion, not from attics or crawlspaces. Believe it or not, outside air is an awful lot cleaner than inside air, and we spend about 90% of our time inside.

After sealing all of my downlights by enclosing them in the attic with airtight boxes made of foam board, Ori fired up the blower door again and measured 6.4 ACH at 50 Pa, or 0.4 ACH by natural ventilation—not perfect, but much better than before. To tighten up my 56-year-old house even more would require more time and effort than would be cost-effective. Ori and crew then sealed all the ductwork in the attic and blew in several inches of cellulose insulation. Now my wife and I are breathing easier and ready to be snug in our house this winter. And now that we won't have to heat the attic, our heating bills will be about half what they would have been without the retrofit.

—Jim Gunshinan

*Jim Gunshinan is **Home Energy's** managing editor.*

*For more information on recessed-can lights, go to "A Recessed Can of Worms," **HE** Jan/Feb 01, p. 42; and "Further Wrestling with Recessed-Can Lights," **HE** Sept/Oct '05, p. 30.*

in general are addressing their clients' wishes for home comfort, health, and energy savings (see "From HVAC to Home Performance Contractor," *HE*, Mar/Apr '07, p. 8). Many who already have the truck, the contacts, and some of the tools see it as a natural evolution in their business, and many educational opportunities are popping up all over the nation to get them started (see "Training Guide for Home Performance Professionals," *HE* July/Aug '07, p. 24).

Training and More Training

The investment in home performance contracting, however, involves more than a weekend crash course. One of the barriers to entry

into this emerging field is the need for extensive initial and some ongoing staff training. It can be very well worthwhile, but a typical contractor may be reticent to take top fieldworkers away from their jobs for a week of training.

The training that launched AHE into home performance was the six-day course taught by the CBPCA mentioned above. It's a free course taught at Pacific Gas and Electric Company (PG&E)'s Pacific Training Center, in Stockton, California. This program is funded by the California Public Utilities Commission (from customers' utility bills) and is supported by Energy Star's Home Performance with Energy Star program. CBPCA offers training and certification



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Jim Gunshinan checks out the source of significant air leakage in his home—recessed-can lights. When these were sealed the whole house became significantly tighter.

in San Francisco, and CBPCA will loan out equipment when it's available.

All of AHE's new employees first go into the field to get some hands-on experience with the work. If they show interest and ability, then Skloot will train them himself until there is a professional training class available. This may also be a great way to stay on top of the latest tools and methods. The worthiness of the investment is in the comfortable clients and—oh, by the way—environmental benefits and energy savings.

As for the future, the long-range plan for Advanced Home Energy is to ride the wave of demand and visibility that home performance is starting to experience. "We want to keep doing what we're doing but do it on a larger scale," Skloot says. "We plan to approach the realtor community to include home performance at time of sale. In this market that makes sense because home buyers are much more interested and concerned about the monthly costs of home ownership. Everyone is aware that utility costs are only going to increase and that a home

with a proven track record of energy efficiency is a better buy than a typical comparable home." Skloot would also like to offer more services while AHE crew are in the home. Once they are there, and have the client's confidence, there are many green areas that they can address. Examples include water consumption (graywater systems, dual-flush toilets, permaculture, and so on) and remodeling using green materials. "There is so much potential for growth in this field; now we are constantly looking for synergies with other contractors where we can partner and both benefit. I have found that this takes educating other contractors about the importance of what we do and sometimes changing the way they do

to contractors, helps them adapt their business to this new service, and monitors the quality of their work. The training, literature and post-course follow-up with students is good and is only improving as more and more professionals catch this wave. Upon completing the course, contractors have to sign a participation agreement and pay a small fee to become members of CBPCA and be able to promote themselves as Home Performance with Energy Star contractors. They are then allowed to display the Energy Star logo on their promotional materials, and receive promotional boosts from CBPCA. There is a tool library at PG&E's Pacific Energy Training Center

their work," says Skloot. For example, AHE works with roofers to make sure that the home is insulated correctly and prepped for solar.

For Ori Skloot, AHE is a smart and scalable model of environmental activism. It is a particularly good model, because rather than needing to fund-raise in order to spread the word for the environment and conservation, AHE offers a valuable service, makes a profit and is able to expand its scope and services. Does social justice also come into the picture? Skloot had to think about that question for a moment, and yet there he was, assisted by two workers (with whom he was speaking fluent Spanish), and right in the work with them. AHE also walks its talk; its trucks run on biodiesel, staff vehicles are hybrids, and it builds with recycled lumber, to name just a few of its many sustainable practices—democracy in action without fanfare.

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Leslie Jackson is the new associate editor at Home Energy.

For more information:

Advanced Home Energy

1013 Pardee St., Ste. 212
Berkeley, CA 94710
Tel: (510)540-4860
Web site:
www.advancedhomeenergy.com

California Building Performance Contractors Association (CBPCA)

Tel: (888)352-2722 (Northern and Central California); (888)357-1777 (Southern California)
Web site: www.cbPCA.org

Home Performance with Energy Star

Web site: www.energystar.gov

Pacific Energy Center, Stockton

Web site: www.pge.com/education_training/classes/energy_efficiency

Rising Sun Energy Center

Web site: www.risingsunenergy.org